

The Optimization Coursebook

Everything md2textbook can do, written in plain
Markdown

md2textbook
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Foundations

1.1 Why this chapter exists

LEARNING OBJECTIVES

After this tour you will know how each Markdown construct becomes a page element:

- **callout boxes** in a dozen semantic colors,
- **equations** that look like they came from a paper,
- **code listings** with highlighting, line numbers and captions,
- **tables, figures and cross-chapter numbering** that stay consistent.

This document is the demo. Every box, equation, table and listing below is ordinary Markdown, converted offline with one command. Headings become numbered chapters and sections, the contents page builds itself, and every heading is a clickable PDF bookmark.

Inline formatting stays restrained: **bold** for emphasis, *italics* for terms, **inline code** in a warm accent, [links](#) in indigo, and math such as $f(x) = \sigma(w^T x + b)$ set in a serif italic that matches the body.

1.1.1 The problem in one line

Learning means choosing parameters $\theta \in \mathbb{R}^d$ that make a loss small:

$$\theta^\star = \arg \min_{\theta} L(\theta), \quad L(\theta) = \frac{1}{N} \sum_{i=1}^N \ell(f_{\theta}(x_i), y_i) \quad (1.1)$$

DEFINITION

Gradient

The vector $\nabla L(\theta) = \left(\frac{\partial L}{\partial \theta_1}, \dots, \frac{\partial L}{\partial \theta_d} \right)$ of partial derivatives. It points in the direction of steepest *increase* of L .

WORKED EXAMPLE

A worked step

Take $L(\theta) = \theta^2$, start at $\theta_0 = 4$ and use a learning rate $\eta = 0.25$. The gradient is $2\theta_0 = 8$, so $\theta_1 = 4 - 0.25 \cdot 8 = 2$. One more step gives $\theta_2 = 2 - 0.25 \cdot 4 = 1$: the value halves every time.

INSIGHT

Why halving?

Each update multiplies θ by $(1 - 2\eta)$. For $\eta = 0.25$ that factor is 0.5, so convergence is geometric. At $\eta = 1$ the factor is -1 and the iterate bounces between two values forever.

WARNING

For this quadratic, any $\eta > 1$ makes the factor $|1 - 2\eta| > 1$ and the iteration **diverges**.

1.2 The update rule

1.2.1 Gradient descent

Starting from θ_0 , repeat

$$\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t) \quad (1.2)$$

Why does stepping against the gradient help? A first-order expansion shows the loss change for a small step δ :

$$L(\theta + \delta) \approx L(\theta) + \nabla L(\theta)^\top \delta \quad (1.3)$$

$$\delta^\star = -\eta \nabla L(\theta) \quad (1.4)$$

$$L(\theta + \delta^\star) \approx L(\theta) - \eta \|\nabla L(\theta)\|^2 \quad (1.5)$$

Equations are numbered per chapter, like (1.1) and (1.2), so you can refer to them from the text.

1.2.2 Momentum and Adam

Momentum accumulates a velocity $v_t = \beta v_{t-1} + \nabla L(\theta_t)$. Adam also tracks a second moment and divides by its square root:

$$\theta_{t+1} = \theta_t - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}} \quad (1.6)$$

THEOREM**Convergence on convex quadratics**

If $L(\theta) = \frac{1}{2} \theta^\top A \theta$ with eigenvalues in $[\mu, L_{\max}]$, gradient descent with $\eta = \frac{2}{\mu + L_{\max}}$ satisfies

$$\|\theta_t - \theta^\star\| \leq \left(\frac{\kappa - 1}{\kappa + 1}\right)^t \|\theta_0 - \theta^\star\|, \quad \kappa = \frac{L_{\max}}{\mu}.$$

TIP

Plot the loss on a log axis. Straight lines mean geometric convergence; a flat tail means you have hit the noise floor.

Code and Data

2.1 Listings

Fenced code is highlighted with an editor-style dark theme. Add `title="..."` to get a numbered caption, and long lines wrap with a continuation arrow instead of shrinking.

Listing 2.1 Gradient descent with momentum

```
1  import numpy as np
2
3
4  def descend(grad, theta0, lr=0.1, beta=0.9, steps=200):
5      """Minimise a function given its gradient."""
6      theta, velocity = np.asarray(theta0, dtype=float), 0.0
7      history = [theta.copy()]
8      for step in range(steps):
9          velocity = beta * velocity + grad(theta)
10         theta = theta - lr * velocity # one update
11         history.append(theta.copy())
12     return theta, np.array(history)
13
14
15 theta, path = descend(lambda t: np.array([1.0, 5.0]) * t, [2.6, 2.2], lr=0.05)
16 print(f"final point: {theta} after {len(path) - 1} steps")
```

Other languages work the same way:

Listing 2.2 The same update in Rust

```
1  fn step(theta: &mut [f64], grad: &[f64], lr: f64) {
2      for (t, g) in theta.iter_mut().zip(grad) {
3          *t -= lr * g;
4      }
5  }
```

Listing 2.3 A training configuration

```
1  optimizer:
2      name: adam
3      lr: 3.0e-4
4      betas: [0.9, 0.999]
5  schedule:
6      type: cosine
7      warmup_steps: 500
```

Shell snippets are shown without line numbers:

```
pipx install md2textbook
md2textbook showcase.md --no-open
```

2.2 Tables

Pipe tables get a dark header, zebra rows, alignment from the separator row, and a repeating header if they run onto another page.

Table 2.1 Common optimizers compared

Optimizer	Memory	Typical learning rate	Notes
SGD	$O(d)$	0.1	Simple and strong with a good schedule
SGD + momentum	$O(d)$	0.05	Smooths noisy gradients
Adam	$O(2d)$	0.0003	Default for transformers; needs warmup
L-BFGS	$O(md)$	1.0	Full batch only; great for small problems

2.3 Lists

- Bullets nest to any depth
 - second level
 - third level
- Task lists are drawn as check boxes:
 - ☒ write the Markdown
 - ☒ run `md2textbook`
 - ☐ enjoy the PDF
- 1. Numbered lists keep their numbering
- 2. including nested items
 1. like this one
 2. and this one
- 3. and carry on afterwards

QUICK REFERENCE

Cheat sheet

- Start a box with `> [!KIND]` or `> **Kind:**`
- Kinds include DEFINITION, EXAMPLE, INSIGHT, WARNING, RECAP, OBJECTIVE, CODE, IDEA, REVIEW, BRIEF, MATH and NOTE
- Equations: `$inline$` and `$$display$$`
- Code: add `title="..."` for a captioned listing

Figures and Boxes

3.1 Figures

Images keep their aspect ratio, never overflow the page, and get a numbered caption taken from the alt text.

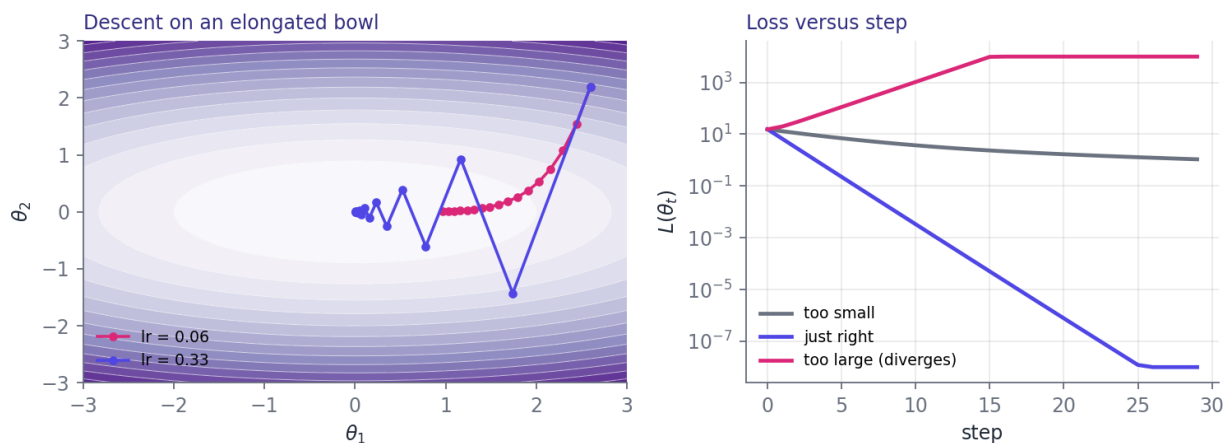


Figure 3.1 Gradient descent on an elongated bowl: two learning rates (left) and loss curves for three step sizes (right)

3.2 The full set of boxes

NOTE

A note: neutral background information that is useful but not essential.

CODE

Implementation hint

Keep the learning-rate schedule in one function so experiments only change a single line.

RESEARCH IDEA

Research idea

Adapt η per layer from the ratio of weight norm to gradient norm, and test whether it removes the need for warmup.

REVIEWER'S VIEW

Reviewer's view

The convergence claim holds only for convex quadratics. The text should say what changes for non-convex losses.

MATH**Derivation**

Setting the derivative of $L(\theta + \delta) = L(\theta) + g^T \delta + \frac{1}{2\eta} \|\delta\|^2$ to zero gives $\delta = -\eta g$, the gradient step.

RECAP**Key takeaways**

- The update rule is one line: $\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t)$.
- The learning rate decides everything: too small is slow, too large diverges.
- Momentum and Adam help when curvature differs across directions.

A plain blockquote becomes a quiet, unlabeled note:

"The art of being wise is the art of knowing what to overlook." *William James*

Unicode works without ceremony: α , β , γ , ∇ , Σ , $\leq$, $\approx$, $\neg$ and $\checkmark$ are routed to a font that has them.